

Generation Interconnection Request System Impact Study Report

For Queue Project Q19-1226

Valley View 138 kV
November 02, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1226. The Interconnection Customer proposes a 144.5 MW solar photovoltaic facility interconnecting at the Valley View 138 kV substation in the West Virginia area. The request was received on June 14, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 409 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 13 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$44,881,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Hustisford-Recycling Coors 138 KV line	Pence-Valemount 345 KV Line	135.0	182	16.4
Vly View-Red Maple 138 KV Line	Valemount-Tristate 345 KV	134.1	229	70.4
Tackett Crk-Red Maple 138 KV	Hurricane Substation-N Madison 345 138 KV	132.8	219	42.6
Peaceable-Recycling Coors 138 KV	Pence-Valemount 345 KV	123.7	209	16.4
W Barstow-Hustisford 138 KV	Pence-Valemount Substation 345 KV	123.5	191	16.4
Greenleaf-Peaceable 138 KV	Pence-Valemount 345 KV L	123.0	214	16.4
Greenleaf-Tackett Crk 138 KV line	Valemount-Pence 345 KV L	119.6	231	16.4
Hampshire Paper-Rockaway Beach 138 KV	Poomac Yard-Union Valley 138 KV	114.8	303	6.1
Mattoon East-Rockaway Beach 138 KV	Potomac Yard-Union Vly Substation 138 KV	114.4	307	6.1
N Madison 345-Hurricane 138 KV	Tackett Crk-Red Maple 138 KV	106.0	279	7.9
Union Valley-Potomac Yard 138 KV	Mattoon E-Rockawa Beach 138 KV Line	105.2	251	6.0
Dalewood-Turner 138 KV	Valemount-Pence 345 KV	103.5	212	13.7
Tackett Crk-Dalewood 138 KV	Pence-Valemount 345 KV	101.8	223	13.7

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Recycling Coors-Hustisford 138 KV	Rebuild line with higher-capacity conductor	\$7,612,000	100% (\$7,612,000)
Vly View-Red Maple 138 KV Line	Rebuild line with higher-capacity conductor	\$2,166,000	100% (\$2,166,000)
Red Maple-Tackett Creek 138 KV	Rebuild line with higher-capacity conductor	\$2,619,000	100% (\$2,619,000)
Peaceable-Recycling Coors 138 KV	Rebuild line with higher-capacity conductor	\$2,149,000	100% (\$2,149,000)
W Barstow-Hustisford 138 KV	Rebuild line with higher-capacity conductor	\$1,631,000	100% (\$1,631,000)
Greenleaf-Peaceable 138 KV	Rebuild line with higher-capacity conductor	\$1,571,000	100% (\$1,571,000)
Greenleaf-Tackett Crk 138 KV line	Rebuild line with higher-capacity conductor	\$5,447,000	100% (\$5,447,000)
Rockaway Beach-Hampshire Paper 138 KV	Rebuild line with higher-capacity conductor	\$12,739,000	74% (\$9,448,000)
Rockaway Beach Substation-Mattoon East Substation 138 KV	Rebuild line with higher-capacity conductor	\$5,426,000	74% (\$4,024,000)
N Madison 345-Hurricane 138 KV	Reconductor line	\$784,000	100% (\$784,000)
Union Valley Substation-Potomac Yard 138 KV	Reconductor line	\$4,140,000	100% (\$4,140,000)
Dalewood-Turner 138 KV	Reconductor line	\$1,463,000	100% (\$1,463,000)
Tackett Crk-Dalewood 138 KV	Reconductor line	\$1,827,000	100% (\$1,827,000)
	Total allocated to project		\$44,881,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.